

# FBISE Class 9 Chemistry

Complete solved questions • MCQs, Section B and Section C • Including every OR part

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A classroom handout based on the supplied question images. The original option order is preserved. Where the question itself has an ambiguity or a printing error, the explanation flags it clearly. Some illustrations and formula symbols are typeset rather than copied from the source images.

| Contents  | Questions                                                         |
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| Section A | MCQs (i)–(xii): correct options and why each alternative is wrong |
| Section B | Short questions (i)–(x), including every OR                       |
| Section C | Long questions Q.3–Q.5, including every OR and structures/graphs  |

## Section A — Multiple choice questions

### (i) Primary focus of organic chemistry

Answer: B — The study of carbon-based molecules and their reactions

Organic chemistry mainly studies carbon compounds, including their structures, properties and reactions; methane ( $\text{CH}_4$ ) is an example. Some carbon compounds, such as  $\text{CO}_2$  and carbonates, are conventionally classed as inorganic.

- A. Behavior of inorganic salts belongs mainly to inorganic chemistry.
- B. Correct: carbon-based molecules and their reactions are the primary focus.
- C. Metals and alloys are mainly covered by inorganic chemistry/materials science.
- D. Gases and their interactions are studied across multiple branches and do not define organic chemistry.

### (ii) Largest atomic size: fluorine, oxygen, chlorine or bromine

Answer: D — Bromine

Atomic size increases down a group because each element has another occupied electron shell. Bromine is below chlorine and fluorine in Group 17 and has the largest atom among the choices.

- A. Fluorine has only two occupied shells and is small.
- B. Oxygen also has two occupied shells; it is larger than fluorine in Period 2, but smaller than bromine.
- C. Chlorine is larger than fluorine, but smaller than bromine.
- D. Correct: bromine has the greatest number of occupied shells among these choices.

### (iii) Identify the covalent compound

Answer: C —  $\text{H}_2\text{O}$

Hydrogen and oxygen are nonmetals and share electron pairs in water; their bonds are covalent.

- A.  $\text{MgO}$  is predominantly ionic: a metal (Mg) combines with a nonmetal (O).
- B.  $\text{CaO}$  is predominantly ionic: calcium forms  $\text{Ca}^{2+}$  and oxygen forms  $\text{O}^{2-}$ .
- C. Correct:  $\text{H}_2\text{O}$  contains covalent O–H bonds.

D. KF is ionic: potassium forms  $K^+$  and fluorine forms  $F^-$ .

#### (iv) Reagents for reduction of alkyl halides

Answer: D —  $Zn + HCl$

In the school-level method, zinc and hydrochloric acid provide reducing hydrogen that replaces the halogen in an alkyl halide.  $R-X + 2[H] \rightarrow R-H + HX$ ; R-H is an alkane.

A.  $Al_2O_3$  at 35 °C is not the indicated reducing reagent.

B. Concentrated  $H_2SO_4$  at 170 °C is commonly used to dehydrate an alcohol to an alkene.

C. Zinc dust alone is not the intended reagent pair in this question.

D. Correct:  $Zn + HCl$  supplies reducing hydrogen.

#### (v) Oxidized element in $H_2S + Cl_2 \rightarrow 2HCl + S$

Answer: C — Sulfur (S)

Sulfur goes from  $-2$  in  $H_2S$  to  $0$  as elemental S: an increase in oxidation state is oxidation. Chlorine goes from  $0$  in  $Cl_2$  to  $-1$  in  $HCl$ : it is reduced. Hydrogen stays at  $+1$ .

A. H does not change oxidation state.

B. Cl decreases from  $0$  to  $-1$ , so it is reduced.

C. Correct: S increases from  $-2$  to  $0$ .

D. S is oxidized, but  $Cl_2$  is reduced; both are not oxidized.

#### (vi) Why is proton number unique to each element?

Answer: B — Protons define the atomic number of the element

Atomic number equals proton number. Every carbon atom has six protons; seven protons identify nitrogen. The periodic table is arranged by increasing atomic number.

A. Protons and neutrons both contribute most of the atom's mass; isotopes of one element differ in neutron number.

B. Correct: proton number defines atomic number and the identity of an element.

C. Protons do not fix neutron number; isotopes differ in neutron number.

D. A neutral atom has as many electrons as protons, but an ion can gain or lose electrons without changing element.

#### (vii) Pair of one-mole samples with the same mass

Answer: A — 1 mol  $N_2O$  and 1 mol  $CO_2$

$M(N_2O) = 2(14) + 16 = 44$  g/mol.  $M(CO_2) = 12 + 2(16) = 44$  g/mol. Thus one mole of each weighs 44 g.

A. Correct: both molar masses equal 44 g/mol.

B.  $H_2$  is 2 g/mol;  $NH_3$  is 17 g/mol.

C.  $H_2O$  is 18 g/mol;  $H_2O_2$  is 34 g/mol.

D.  $H_2SO_4$  is 98 g/mol;  $HNO_3$  is 63 g/mol.

#### (viii) Effect of an exothermic reaction on the surroundings

Answer: C — Temperature increases

An exothermic reaction releases heat into the surroundings, making them warmer under ordinary conditions.

A. A decrease would be expected when the surroundings lose heat to an endothermic process.

- B. No change is not the usual effect when heat is released to surroundings.
- C. Correct: surroundings usually get warmer.
- D. Heat release does not make the surroundings' temperature zero.

### (ix) Relative atomic mass of chlorine

Answer: B — 35.5

With 75%  $^{35}\text{Cl}$  and 25%  $^{37}\text{Cl}$ :  $A_r = (35 \times 0.75) + (37 \times 0.25) = 26.25 + 9.25 = 35.5$ . The abundant isotope carries more weight in the average.

- A. 36.0 is the unweighted average and would require equal abundances.
- B. Correct: 35.5 is the weighted average.
- C. 37.0 ignores the more abundant  $^{35}\text{Cl}$ .
- D. 34.5 is impossible as an average of 35 and 37; it is below both.

### (x) Purpose of catalytic converters in vehicles

Answer: B — To reduce harmful gases in exhaust

Catalysts help convert harmful exhaust substances: carbon monoxide ( $\text{CO}$ ) to carbon dioxide ( $\text{CO}_2$ ), and nitrogen oxides to nitrogen, for example.

- A. Increasing fuel consumption is not their purpose.
- B. Correct: they reduce harmful exhaust emissions.
- C. They are not installed to increase vehicle speed.
- D. Sulfur dioxide is a pollutant; adding it would worsen pollution.

### (xi) Compound associated with acid-rain damage to marble

Answer: A —  $\text{CaSO}_4$  (intended answer)

Marble is mainly  $\text{CaCO}_3$ . Sulfuric acid in acid rain attacks it:  $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$ . The marble surface is consumed and calcium sulfate can form a damaging crust.

- A. Correct intended choice:  $\text{CaSO}_4$  is the sulfate formed by sulfuric acid attack.
- B.  $\text{Ca}(\text{NO}_3)_2$  is also a possible product if nitric acid in acid rain reacts with marble. The wording is therefore somewhat ambiguous.
- C.  $\text{CaCO}_3$  is the original main constituent of marble, not the sulfate reaction product.
- D.  $\text{CaC}_2\text{O}_4$  is not the usual product in the stated sulfuric-acid explanation.

### (xii) INCORRECT statement about solubility

Answer: D — It is independent of temperature

Solubility generally changes with temperature. Many solids dissolve more in hotter water; gases generally become less soluble as temperature rises.

- A. A basic description: a solute's ability to dissolve, though a precise definition states a maximum quantity under stated conditions.
- B. Grams per liter of solvent is one possible expression, alongside grams per 100 g of solvent.
- C. True in general: temperature and the nature of solute/solvent matter; pressure matters most for gas solubility.
- D. Incorrect, and thus the answer: solubility is not generally independent of temperature.

## Section B — Short questions and OR parts

### B(i) Bond breaking and bond making

- a. Bond breaking requires an input of energy to overcome attraction between bonded atoms. It is an endothermic process.
- b. Bond breaking absorbs energy, whereas bond making releases energy (an exothermic process).

#### OR — Identify functional groups

| Compound                           | Group | Name                       |
|------------------------------------|-------|----------------------------|
| CH <sub>3</sub> COOH               | -COOH | Carboxyl (carboxylic acid) |
| CH <sub>3</sub> CHO                | -CHO  | Aldehyde                   |
| CH <sub>3</sub> CH <sub>2</sub> OH | -OH   | Hydroxyl (alcohol)         |

### B(ii) Reaction of chlorine with halide ions

Chlorine is more reactive than bromine and iodine, so it displaces them from their halide ions.

- a. Bromide:  $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$ . Bromide is oxidized to bromine.
- b. Iodide:  $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$ . Iodide is oxidized to iodine. The coefficient 2 balances atoms and charge.

#### OR — Protons and neutrons

For a nuclide  ${}^A_Z\text{E}$ , protons = Z and neutrons = A – Z.

| Nuclide                | Protons | Neutrons | Working        |
|------------------------|---------|----------|----------------|
| ${}^{27}_{13}\text{X}$ | 13      | 14       | $27 - 13 = 14$ |
| ${}^{36}_{18}\text{Y}$ | 18      | 18       | $36 - 18 = 18$ |
| ${}^{64}_{30}\text{Z}$ | 30      | 34       | $64 - 30 = 34$ |

### B(iii) Liquid crystals versus ordinary liquids

| Property              | Liquid crystals                                                       | Ordinary liquids                                                          |
|-----------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------|
| Molecular arrangement | Have some order, often a preferred molecular orientation.             | Have no sustained preferred molecular orientation; arrangement is random. |
| Fluidity              | Flow, but flow may depend on direction of alignment and be less free. | Flow freely and take the shape of the container.                          |

Liquid crystals combine partial molecular order with an ability to flow.

#### OR — Oxidizing agents

An oxidizing agent causes another substance to be oxidized and is itself reduced.

- a.  $2\text{NH}_3 + 3\text{CuO} \rightarrow 3\text{Cu} + \text{N}_2 + 3\text{H}_2\text{O}$ . CuO is the oxidizing agent: Cu changes from +2 in CuO to 0 in Cu. N changes from –3 in NH<sub>3</sub> to 0 in N<sub>2</sub>.

b.  $2\text{FeCl}_2 + \text{Cl}_2 \rightarrow 2\text{FeCl}_3$ .  $\text{Cl}_2$  is the oxidizing agent: Cl changes from 0 to  $-1$ ; Fe changes from  $+2$  to  $+3$ .

### B(iv) Effects of water pollutants on organisms

1. Household waste: Sewage can carry disease-causing microorganisms; decomposition also consumes dissolved oxygen and can kill aquatic life.
2. Agricultural waste: Fertilizer runoff promotes excess algal growth; decomposition depletes oxygen. Pesticides can poison aquatic organisms.
3. Metals: Toxic metals such as lead and mercury accumulate in organisms and can damage organs and the nervous system.

#### OR — Sunlight in chlorination of methane

Sunlight (particularly UV light) supplies energy to split  $\text{Cl}_2$  into reactive chlorine radicals:  $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$ . These start substitution. One hydrogen in methane is replaced by chlorine:



Products: chloromethane and hydrogen chloride. Further substitution can occur on continued chlorination.

### B(v) Transition metals as catalysts

| Metal         | Catalytic use                                      |
|---------------|----------------------------------------------------|
| Iron (Fe)     | Haber process: synthesis of ammonia.               |
| Nickel (Ni)   | Hydrogenation of vegetable oils.                   |
| Platinum (Pt) | Catalytic converter: treatment of vehicle exhaust. |

A catalyst increases the rate of a reaction without being permanently consumed.

#### OR — Nutrients supplied by fertilizers

Urea mainly supplies nitrogen; ammonium salts supply nitrogen as  $\text{NH}_4^+$ ; nitrates supply nitrogen as  $\text{NO}_3^-$ . The primary nutrient asked for is nitrogen in each case; only its chemical form differs.

### B(vi) Molecular and structural formulas of $\text{C}_4\text{H}_{10}$

A molecular formula gives the types and numbers of atoms, while a structural formula shows which atoms are bonded to which. Both isomers have molecular formula  $\text{C}_4\text{H}_{10}$ :

| Isomer          | Condensed structural formula                         | Arrangement           |
|-----------------|------------------------------------------------------|-----------------------|
| Butane          | $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CH}_3$ | Straight carbon chain |
| 2-Methylpropane | $\text{CH}_3\text{--CH(CH}_3\text{)--CH}_3$          | Branched carbon chain |

The same molecular formula does not distinguish their different connections; the structural formulas do.

#### OR — Group and period from nuclide symbols

| Nuclide                  | Electron arrangement | Group | Period |
|--------------------------|----------------------|-------|--------|
| $^{12}_6\text{A (C)}$    | 2, 4                 | 14    | 2      |
| $^{39}_{19}\text{B (K)}$ | 2, 8, 8, 1           | 1     | 4      |
| $^{19}_9\text{C (F)}$    | 2, 7                 | 17    | 2      |

The lower number gives the atomic number. Occupied shells determine the period; outer-shell electrons identify the group for these main-group elements.

### B(vii) Shifts in the cobalt chloride hydrate equilibrium

The printed equation shows  $\text{CoCl}_2 \cdot 2\text{H}_2\text{O(s)} + \text{heat} \rightleftharpoons \text{CoCl}_2(\text{g}) + \text{H}_2\text{O(g)}$ . It is unbalanced and gives an inappropriate gas state for  $\text{CoCl}_2$ . The corrected dehydration expression is:



| Change               | Shift | Reason                                                               |
|----------------------|-------|----------------------------------------------------------------------|
| a. Heating           | Right | Heat is on reactant side, favouring the endothermic forward process. |
| b. Addition of water | Left  | More product water favours rehydration / the reverse process.        |
| c. Removal of water  | Right | The system favours production of replacement water.                  |

These shifts assume that adding or removing water changes the water vapour available to this equilibrium; Le Chatelier's principle predicts the response.

### OR — Carbohydrates as an energy source

Carbohydrates are digested into glucose, which cells can use readily in respiration. They are an important quick source of usable energy; proteins primarily support growth and repair, while fat often provides stored, longer-term energy.

Scientific qualification: "More efficient" should mean more readily available for quick use. Fat is more energy-dense: roughly 9 kcal/g versus about 4 kcal/g for carbohydrates and proteins. Thus carbohydrates do not supply more energy per gram than fats.

### B(viii) Closed systems and dynamic equilibrium

A closed system prevents reactants or products from leaving or entering, so both remain available. Initially the forward reaction makes products; as products accumulate, the reverse reaction increases. When the forward and reverse rates become equal, dynamic equilibrium is established: reactions continue, but concentrations remain constant.

### OR — Brønsted-Lowry acids and bases

A Brønsted-Lowry acid donates a proton ( $\text{H}^+$ ); examples:  $\text{HCl}$  and  $\text{H}_2\text{SO}_4$ . A Brønsted-Lowry base accepts a proton; examples:  $\text{NH}_3$  and  $\text{OH}^-$ . Example:  $\text{HCl} + \text{NH}_3 \rightarrow \text{NH}_4^+ + \text{Cl}^-$ :  $\text{HCl}$  donates and  $\text{NH}_3$  accepts  $\text{H}^+$ .

### B(ix) Calculate numbers of atoms

Use Avogadro's constant  $N_A = 6.022 \times 10^{23} \text{ particles mol}^{-1}$ .

a. 3.4 mol  $N_2$ : each molecule has two N atoms; number of N atoms =  $3.4 \times 2 \times 6.022 \times 10^{23} = 4.09 \times 10^{24}$  atoms.

b. 15 g Na: moles Na = 15/23 mol; number of Na atoms =  $(15/23) \times 6.022 \times 10^{23} = 3.93 \times 10^{23}$  atoms.

### OR — Endothermic versus exothermic energy profiles

| Feature                    | Endothermic              | Exothermic                |
|----------------------------|--------------------------|---------------------------|
| Product energy             | Above reactant level     | Below reactant level      |
| Enthalpy change $\Delta H$ | Positive; absorbs energy | Negative; releases energy |
| Forward activation energy  | Reactant level to peak   | Reactant level to peak    |

Both require activation energy. In two opposite directions of the same reaction, the endothermic direction has the larger activation energy. Across unrelated reactions, one cannot assume that every endothermic reaction has a larger  $E_a$  than every exothermic one.

### B(x) Isotopes of hydrogen

Isotopes are atoms of one element with the same proton number (atomic number Z) but different neutron numbers and therefore different mass numbers A.

| Isotope   | Symbol | Z (protons) | A | Neutrons |
|-----------|--------|-------------|---|----------|
| Protium   | $^1H$  | 1           | 1 | 0        |
| Deuterium | $^2H$  | 1           | 2 | 1        |
| Tritium   | $^3H$  | 1           | 3 | 2        |

### OR — Branches of chemistry

Astrochemistry: chemical substances and reactions in space (stars, planets and interstellar clouds).

Geochemistry: the composition and chemical processes of Earth, including minerals, rocks, soils and water.

Physical chemistry: physical principles governing substances and reactions, including energy changes, rates and equilibrium.

## Section C — Long questions and OR parts

### Q.3 — Periodic trends in Group IIA (Group 2)

Group IIA contains Be, Mg, Ca, Sr and Ba. Each has two valence electrons. Moving down this group adds an occupied shell at every step.

| Property                | Trend from Be to Ba | Explanation                                                                     |
|-------------------------|---------------------|---------------------------------------------------------------------------------|
| Atomic size             | Increases           | Additional occupied shells place the outer electrons farther from the nucleus.  |
| First ionization energy | Decreases           | Greater distance and shielding weaken nuclear attraction to the outer electron. |
| Electronegativity       | Generally decreases | Larger atoms with more shielding attract a shared bonding pair less strongly.   |

**Down Group IIA: size ↑ ionization energy ↓ electronegativity ↓**

#### Q.3 OR — Dot-and-cross and Lewis structures

Drawing rule: one shared pair is one covalent bond; two pairs form a double bond; three pairs form a triple bond. Different symbols indicate which atom supplied each electron.

1. HCN (hydrogen cyanide): H-C-N atom sequence. Let ○ = H electron, • = C electron, × = N electron. The H-C bond is (○•); the C≡N bond contains three (•×) pairs; nitrogen has one (××) lone pair.

**Dot-and-cross: H(○•)C(•×)(•×)(•×)N(××)**

**Lewis: H-C≡N:**

2. MgO (magnesium oxide): ionic bonding. Mg loses two electrons, becoming  $\text{Mg}^{2+}$ ; O gains them, becoming  $\text{O}^{2-}$ . The six original outer electrons on O are • and the two donated by Mg are ×. Place both ions in brackets with their charges; O then has four electron pairs.

**Dot-and-cross:  $[\text{Mg}]^{2+}$  [O surrounded by ••, ••, •• and ××] $^{2-}$**

Lewis:  $[\text{Mg}]^{2+}$  [O with four lone pairs] $^{2-}$ . There is no shared Mg-O bonding pair.

3.  $\text{C}_2\text{H}_4$  (ethene): Each C makes two C-H single bonds and a C=C double bond.  $\text{H}_2\text{C}=\text{CH}_2$  is its Lewis structural formula. The double bond contains two shared pairs. Let • = left C, × = right C, ○ = H: each left C-H pair is (•○), each right C-H pair is (×○), and the C=C bond contains two (•×) pairs.

#### LEWIS STRUCTURES (electron pairs indicated where needed)

HCN: H-C≡N:

MgO:  $[\text{Mg}]^{2+}$  [ :O: ] $^{2-}$

$\text{O}^{2-}$  has four lone pairs in all

Ethene:

$$\begin{array}{ccc}
 & \text{H} & \text{H} \\
 & | & | \\
 \text{H}-\text{C} & = & \text{C}-\text{H} \\
 & | & | \\
 & \text{H} & \text{H}
 \end{array}$$

## Q.4 — Reactions of hydrochloric acid

A. With Zn (single displacement):  $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2 \uparrow$ . Zinc displaces hydrogen. Products are zinc chloride and hydrogen gas; gas bubbles appear.

B. With NaOH (neutralization):  $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ . Acid and alkali form salt (sodium chloride) and water.

C. With  $\text{Na}_2\text{CO}_3$  (acid-carbonate reaction):  $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$ . Gas effervescence is seen;  $\text{CO}_2$  turns limewater milky:  $\text{CO}_2 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCO}_3 \downarrow + \text{H}_2\text{O}$ .

## Q.4 OR — Prepare alkanes and name reaction types

| Starting compound                                                | Equation and product                                                                                                                            | Type                                                               |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| A. Propene, $\text{CH}_3\text{-CH=CH}_2$                         | $\text{CH}_3\text{CH=CH}_2 + \text{H}_2 \xrightarrow{\text{Ni}} \text{CH}_3\text{CH}_2\text{CH}_3$ (propane)                                    | Hydrogenation; addition across $\text{C=C}$ .                      |
| B. But-2-yne, $\text{CH}_3\text{-C}\equiv\text{C-CH}_3$          | $\text{CH}_3\text{C}\equiv\text{CCH}_3 + 2\text{H}_2 \xrightarrow{\text{Ni}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (butane)             | Complete hydrogenation; addition across $\text{C}\equiv\text{C}$ . |
| C. 1-Chloropropane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ | $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + 2[\text{H}] \xrightarrow{\text{Zn/HCl}} \text{CH}_3\text{CH}_2\text{CH}_3 + \text{HCl}$ (propane) | Reduction of an alkyl halide; Cl replaced by H.                    |

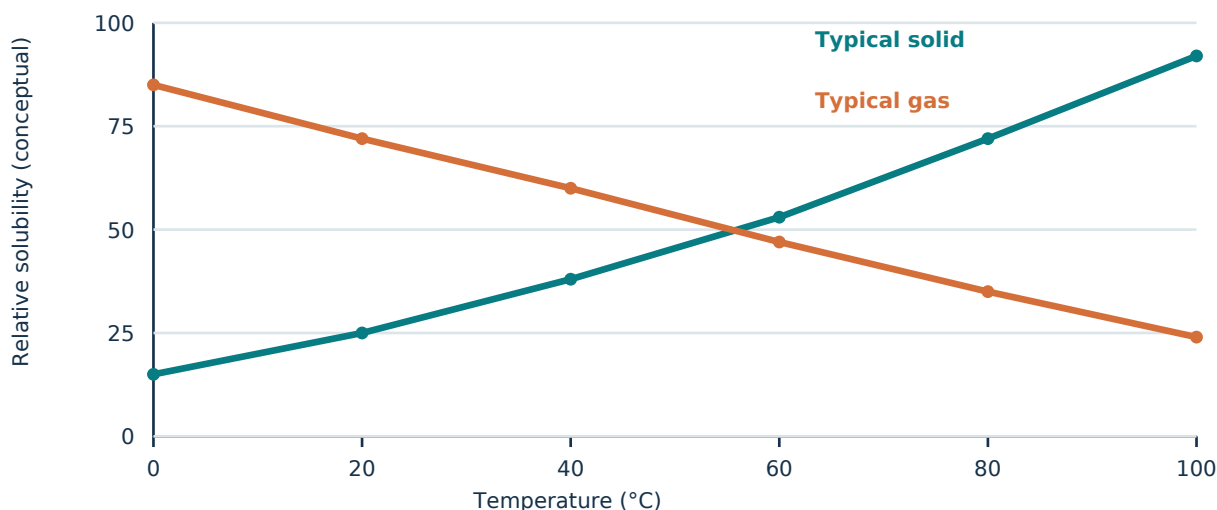
One mole of  $\text{H}_2$  saturates one  $\text{C=C}$  bond; two moles of  $\text{H}_2$  fully saturate one  $\text{C}\equiv\text{C}$  bond.  $[\text{H}]$  represents reducing hydrogen in the school-level alkyl halide reaction.

## Q.5 — Solubility and temperature

Definition: solubility is the maximum amount of solute that can dissolve in a specified amount of solvent at a stated temperature (and pressure where relevant), forming a saturated solution. One expression is grams of solute per 100 g solvent.

Many solids in liquids: solubility rises as temperature rises, so hot solvent can dissolve more of the solid; exceptions exist and depend on the substance. Gases in liquids: solubility usually falls with increasing temperature; warm water generally retains less dissolved gas.

Conceptual trend only — values are illustrative, not measurements



For a hand-drawn graph, label both axes and identify the rising line as a typical solid and the falling line as a typical gas. The diagram represents trends and not actual measured solubilities.

#### **Q.5 OR — Lipid sources and recommended intake**

Sources: plant oils (olive, sunflower), nuts, seeds and avocado; animal foods including butter, ghee, dairy, meat, eggs and oily fish. Lipids include fats and oils.

Intake for young adults: a practical general range is about 20–30% of total daily energy from fat, preferably mostly unsaturated fats. WHO states a minimum of 15% energy from fat for adults and suggests limiting it to 30% or less of total energy to help prevent unhealthy weight gain.

Requirements vary with total calorie needs.

At an illustrative 2,000 kcal/day, 20–30% supplies 400–600 kcal from fat; dividing by approximately 9 kcal/g gives about 44–67 g fat/day. This is an example, not one fixed allowance for everyone.

Source: World Health Organization, “Healthy diet,” 26 January 2026,  
<https://www.who.int/news-room/fact-sheets/detail/healthy-diet>